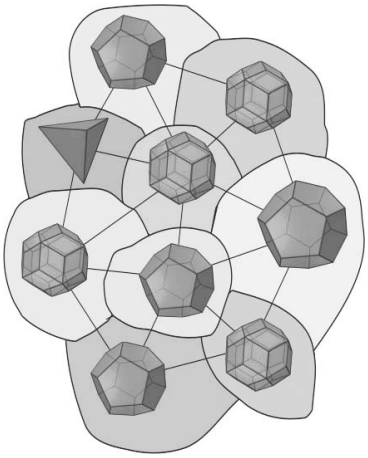
The elementary *quanta* of the gravitational field exist at the Planck scale. They are the elementary grains that weave the mobile fabric with which Einstein reinterpreted Newton's absolute space and time. It is these, and their interactions, that determine the extension of space and the duration of time.

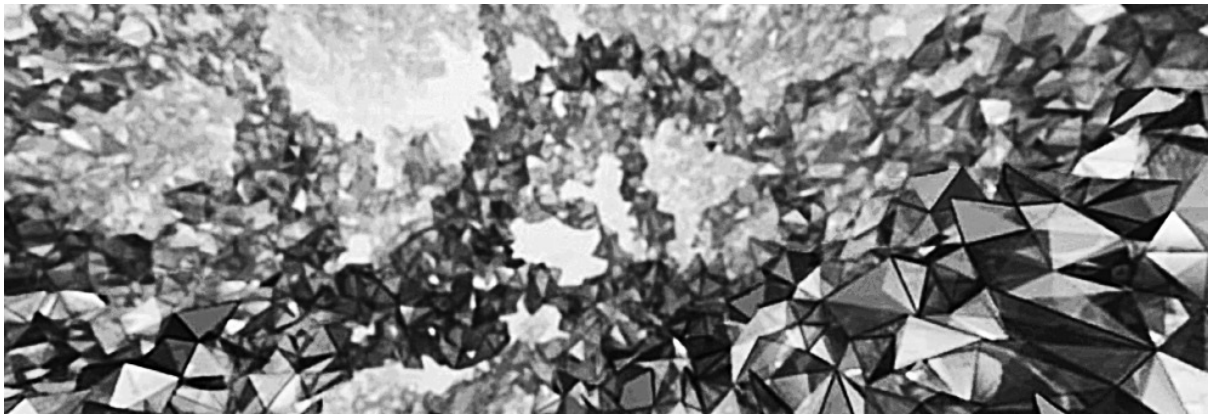


Representation of the web of elementary grains of space (or spin network).

The relations of spatial adjacency tie the grains of space into webs. We call these "spin networks." The name "spin" comes from the mathematics that describe the grains of space.[75] A ring in the spin network is called a "loop," and these are the loops that give "loop theory" its name.

The webs, in turn, transform into each other in discrete leaps, described in the theory as structures called "spinfoam."[76]

The occurrence of these leaps draws the patterns that on a large scale appear to us like the smooth structure of spacetime. On a small scale, the theory describes a "quantum spacetime" that is fluctuating, probabilistic, and discrete. At this scale, there is only the frenzied swarming of quanta that appear and vanish.



Representation of spinfoam.

This is the world with which I seek daily to come to terms. An unusual world, but not a meaningless one.

In my research group in Marseille, for example, we are attempting to calculate the time needed for a black hole to explode when it passes through a quantum phase.